L26: Comparing two proportions

CI for the difference in two proportions
Hypothesis testing – two samples binary data
Two sample hypothesis testing in R

Learning objectives

CI for the difference in two proportions
Hypothesis testing – two samples binary data
Two sample hypothesis testing in R

Last lecture we focused on categorical outcomes (binary/proportions) and covered a one sample test of proportions.

In this lecture we will look at comparing two proportions:

- confidence intervals for the difference between two proportions
- the plus four interval when the samples are small
- hypothesis testing: the two sample test of proportions
- doing all of it in R with `prop.test()`

Recall from last lecture: one proportion

For a single sample, when $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$:

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Two features carry over to today:

- the outcome is binary, so the Normal distribution and z^* do the work, not the t distribution
- the interval is only trustworthy when there are enough successes and enough failures

What changes today is that we have two independent samples, so we need the standard error of a difference.

Comparing two proportions (B & M text Chapter 20)

- Two SRS from independent populations

Notation:

Population	Population proportion	Sample size	Sample proportion
1	p_1	n_1	$\hat{p}_1$
2	p_2	n_2	$\hat{p}_2$

You choose which group is population 1. Swapping the labels flips the sign of the difference and of the interval, and changes nothing else. Say out loud which group you called 1 before you interpret the answer.

CI for the difference in two proportions

Large-sample confidence interval for the difference of two proportions

- Use when the number of observed successes and failures are > 10 for both samples

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

- Just like for the difference between two means, the SE of the difference is the square root of the sum of the variances.
- This large-sample interval often has a lower confidence level than the one specified. That is, if you repeated the method many times, < 95 of the 100 created intervals would contain the true value for the difference between the proportions for a 95% CI.

Example using the large sample method

Patients in a randomized controlled trial were severely immobilized and randomly assigned to either Fragamin (to prevent blood clots) or to placebo. The number of patients experiencing deep vein thrombosis (DVT) was recorded

	DVT	no DVT	Total	$\hat{p}$
Fragamin	42	1476	1518	0.0277
Placebo	73	1400	1473	0.0496

- We can apply the large sample method because the sample sizes are large and the number of observed successes and failures are > 10 (i.e., 42, 73, 1476, and 1400 all > 10).
- Take population 1 = placebo and population 2 = Fragamin, so a positive difference means more clots on placebo.

Example using the large sample method

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

$$(0.0496 - 0.0277) \pm z^* \sqrt{\frac{0.0496(1-0.0496)}{1473} + \frac{0.0277(1-0.0277)}{1518}}$$

$$0.0219 \pm 1.96 \times 0.0071 = 0.008 \text{ to } 0.0358$$

Therefore, the 95% CI is $(0.8\%, 3.6\%)$. Note that the interval excludes 0.

Plus 4 method for the comparison of two proportions

- When the assumptions of the large sample method are not satisfied, we use the plus four method.
- With two samples this method says: add 4 observations in total, 1 success and 1 failure to each of the two samples.

$$\tilde{p}_1 = \frac{\text{no. of successes in pop } 1 + 1}{n_1 + 2} \qquad \tilde{p}_2 = \frac{\text{no. of successes in pop } 2 + 1}{n_2 + 2}$$

$$(\tilde{p}_1 - \tilde{p}_2) \pm z^* \sqrt{\frac{\tilde{p}_1(1-\tilde{p}_1)}{n_1 + 2} + \frac{\tilde{p}_2(1-\tilde{p}_2)}{n_2 + 2}}$$

- Use when each sample size is at least five, with any counts of success and failure (can even use when the number of successes or failures $= 0$)
- Much more accurate when the sample sizes are small
- May be conservative (giving a higher level of confidence than the one specified)

Example using the plus four method

	Flu	no Flu	Total	$\hat{p}$
Vaccine	4	96	100	0.04
Placebo	11	89	100	0.11

Here, we don't have 10 "successes" (flu) in both groups, so we cannot use the Normal approximation method.

Take population 1 = vaccine and population 2 = placebo.

Example using the plus four method

$$\tilde{p}_1 = \frac{4+1}{100+2} = \frac{5}{102} = 0.049 \qquad \tilde{p}_2 = \frac{11+1}{100+2} = \frac{12}{102} = 0.118$$

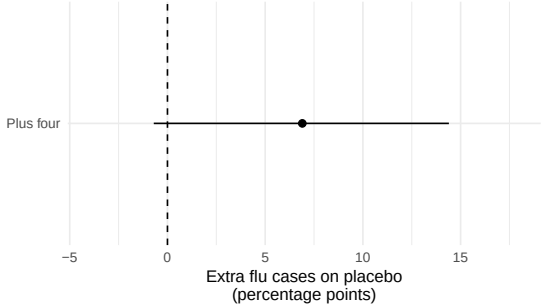
$$(\tilde{p}_2 - \tilde{p}_1) \pm z^* \sqrt{\frac{\tilde{p}_1(1-\tilde{p}_1)}{n_1+2} + \frac{\tilde{p}_2(1-\tilde{p}_2)}{n_2+2}}$$

$$\left(\frac{12}{102} - \frac{5}{102}\right) \pm 1.96 \times 0.0384 = 0.069 \pm 0.075 = -0.007 \text{ to } 0.144$$

Reported as percentage points, the 95% CI runs from -0.7% to 14.4%. Here we put placebo first so that a positive difference means more flu without the vaccine.

Example using the plus four method (continued)

The 95% CI of the difference ranged from -0.7 percentage points to 14.4 percentage points. While this CI contains 0 (the null hypothesized value for no difference) most of the values contained within it are positive, perhaps suggesting support for the alternative hypothesis. In this case, we might want to collect more data to create a more precise CI.



The same interval from R: `prop.test()`

	Flu	no Flu	Total	$\hat{p}$
Vaccine	4	96	100	0.04
Placebo	11	89	100	0.11

```
flu <- c(11, 4)      # successes, placebo first
n   <- c(100, 100)   # sample sizes, same order
prop.test(flu, n)
```

The first argument is the vector of successes, the second is the vector of sample sizes. Keep the two vectors in the same group order.

Reading the `prop.test()` output

```
##
##  2-sample test for equality of proportions with
##  continuity correction
##
## data:  flu out of n
## X-squared = 2.5946, df = 1, p-value = 0.1072
## alternative hypothesis: two.sided
## 95 percent confidence interval:
##  -0.01235958  0.15235958
## sample estimates:
## prop 1 prop 2
##   0.11   0.04
```

Reading the `prop.test()` output

- `prop 1` and `prop 2` confirm the group order, here 0.11 for placebo and 0.04 for vaccine
- the interval, -1.2% to 15.2%, is for $p_1 - p_2$, placebo minus vaccine
- it is a little wider than our plus four interval of -0.7% to 14.4%, because R applies a continuity correction by default
- both intervals contain 0, so both give the same conclusion

R never checks the sample size condition for you. With 4 flu cases in one arm, report the plus four or the R interval, not the large sample one.

Hypothesis testing - two samples binary data

Hypothesis testing when you have two samples and binary data

$H_0 : p_1 = p_2$ or $p_1 - p_2 = 0$

H_a :

- $p_1 \neq p_2$ or $p_1 - p_2 \neq 0$ (two-sided)
- $p_1 > p_2$ or $p_1 - p_2 > 0$ (one sided upper tail)
- $p_1 < p_2$ or $p_1 - p_2 < 0$ (one sided lower tail)

Notice the null names no specific value for p_1 or p_2. It only says the two are equal.

What does it mean to assume the null is true?

- If the null hypothesis is true, then p_1 is truly equal to p_2. In this case, our best estimate of the underlying proportion that they are both equal to is the pooled proportion

$$\hat{p} = \frac{\text{no. successes in both samples}}{\text{no. individuals in both samples}}$$

- Also, our best guess at the SE for the difference is:

$$\sqrt{\frac{\hat{p}(1-\hat{p})}{n_1} + \frac{\hat{p}(1-\hat{p})}{n_2}} = \sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

This is the formula for the SE for the difference between two proportions but we have substituted $\hat{p}$ for p_1 and p_2.

Hypothesis testing when you have two samples and binary data

Using the information from the previous slide, we can create the z-test for the difference between two proportions as:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Use this test when the counts of successes and failures are ≥ 5 in both samples.

The numerator uses the two separate estimates. The denominator uses the pooled estimate, because the test asks what the world would look like if the null were true.

Example of hypothesis testing when you have two samples and binary data

Recall the RCT data on the occurrence of DVT between Fragamin vs. placebo groups:

	DVT	no DVT	Total	$\hat{p}$
Fragamin	42	1476	1518	0.0277
Placebo	73	1400	1473	0.0496

$H_0 : p_1 = p_2$, or that the rate of DVT is the same between Fragamin and placebo groups.

Suppose you're interested in knowing whether these two groups had different rates of DVT. Then, $H_a : p_1 \neq p_2$

Example of hypothesis testing when you have two samples and binary data

1. Compute $\hat{p} = \frac{42+73}{1518+1473} = \frac{115}{2991} = 0.03844868$
2. Compute the SE: $\sqrt{0.0384(1-0.0384)(\frac{1}{1518}+\frac{1}{1473})} = 0.007032308$
3. Compute the test statistic:

$$z = \frac{0.04955872 - 0.02766798}{0.007032308} = 3.11$$

4. Calculate the p-value

```
pnorm(q = 3.112881, lower.tail = FALSE) * 2
```

```
## [1] 0.001852707
```

A difference this large would arise about twice in a thousand trials if the two rates were truly equal.

Two sample hypothesis testing in R

Example of hypothesis testing when you have two samples and binary data

```
prop.test(x = c(42, 73), # x is a vector of number of successes
          n = c(1518, 1473)) # n is a vector of sample sizes
```

```
##
##  2-sample test for equality of proportions with
##  continuity correction
##
## data:  c(42, 73) out of c(1518, 1473)
## X-squared = 9.107, df = 1, p-value = 0.002546
## alternative hypothesis: two.sided
## 95 percent confidence interval:
##  -0.036376917 -0.007404562
## sample estimates:
##     prop 1     prop 2
## 0.02766798 0.04955872
```

Matching the hand calculation

- R gives a slightly different p-value because it has a continuity correction.
- This is okay. If you want to use R to check your hand calculation, you need to add the argument `correct = FALSE` to the calculation.

```
prop.test(x = c(42, 73), n = c(1518, 1473),
          correct = FALSE)$p.value
```

```
## [1] 0.001852706
```

That matches the p-value we computed by hand, 0.0019.

The test and the interval agree

- R reports `X-squared`, not z. A squared standard Normal is a chi-square with 1 degree of freedom, and indeed $3.11^2 = 9.69$
- R put Fragamin first, so its CI of -3.6% to -0.8% is the negative of the interval we built by hand, 0.8% to 3.6%. Same interval, opposite labelling
- the interval excludes 0 and the p-value is below 0.05. For a two-sided test these two statements always travel together

Fragamin lowers the rate of DVT by somewhere between about 1 and 4 percentage points, and the reduction is statistically significant.

Which method should I use?

Situation	Confidence interval	Test
Successes and failures > 10 in both samples	Large sample formula	z test with pooled $\hat{p}$
Small samples, at least 5 per group	Plus four formula	`prop.test()`
In R, any time	`prop.test()`	`prop.test()`

- The CI and the test use different standard errors. The CI uses the two separate proportions, the test uses the pooled one
- Always check the counts before you trust a Normal-based answer
- Always state which group you called 1

Parting Humor

